Homework 06

REAL VARIABLES I

Fall 2026

Due: 2026-10-04 19:00
Topics: The non-negative integral, monotone convergence, Fatou's lemma, and measures defined by densities

Instructions

Submission is through Gradescope, accessible through Blackboard.
Grading rubric: Every problem/subproblem is graded according to this rubric (no intermediate scores):

- **0%**: No submission or no meaningful content.
- **30%**: Attempt at solution, outline of idea.
- **60%**: Serious attempt at solution. Solution fails to address important points.
- **80%**: Solutions with fixable mistakes or incompleteness.
- **100%**: Complete and correct solution.

You may receive up to a 10% penalty for very unclear or messy exposition.
Please start every solution on a separate page. Clearly mark the pages associated to every part. You may submit either computer typeset solutions or handwritten solutions. Handwritten solutions must be *very clearly* legible.
You are encouraged to work in groups. You may submit the entire homework assignment as a group (up to 4 people). If you only did some exercises as a group, instead of a solution you may clearly indicate whose submission I should look at for the solution to the problem.
Indicate any external help/notes/resources you used. You may not use generative AI in any way to help you with the HW.

---

1. (When monotonicity of the integral is strict)

Let $(X, \mathcal{A}, \mu)$ be a measure space, and let $f, g\colon X \to [0, +\infty]$ be measurable. Assume that $g \leq f$ almost everywhere and that

$$\int_X f \, \mathrm{d}\mu < +\infty.$$

1.a

weight: 1.0 Prove the equivalence

$$\int_X f \, \mathrm{d}\mu > \int_X g \, \mathrm{d}\mu \quad \iff \quad \mu(\{x \in X : f(x) > g(x)\}) > 0.$$

Free hint. Since $\int_X f \, \mathrm{d}\mu < +\infty$, the function f is finite almost everywhere. After discarding the null set on which $f = +\infty$ or $g > f$, set $u = f - g$ and use

$$\{u > 0\} = \bigcup_{m=1}^{\infty} \{u \geq 2^{-m}\}.$$

2. (Liminf, limsup, and Fatou's lemma)

Let $(f_n \colon X \to [0, +\infty])_{n \in \mathbb{N}}$ be measurable.

Recall the pointwise definitions

$$\liminf_{n \to \infty} f_n = \sup_{N \in \mathbb{N}} \inf_{n \geq N} f_n, \qquad \limsup_{n \to \infty} f_n = \inf_{N \in \mathbb{N}} \sup_{n \geq N} f_n.$$

For a fixed $x \in X$, explain to yourself, in words or with pictures, what each expression records about the tail values $(f_n(x))_{n \in \mathbb{N}}$.

Make sure you understand that

$$\liminf_{n \to \infty} f_n(x) \leq \limsup_{n \to \infty} f_n(x),$$

and that equality holds if and only if $(f_n(x))$ converges in $[0, +\infty]$.

2.a

weight: 0.7 **(In class.)** Use the monotone convergence theorem and the preceding part to prove Fatou's lemma:

$$\int_X \liminf_{n \to \infty} f_n \, \mathrm{d}\mu \leq \liminf_{n \to \infty} \int_X f_n \, \mathrm{d}\mu.$$

Every step involving an infimum, supremum, or limit should be justified.

> **Free hint.** Define
>
> $$g_N \coloneqq \inf_{n \geq N} f_n.$$
>
> Show that $g_N \uparrow \liminf_n f_n$, apply the monotone convergence theorem, and use $g_N \leq f_n$ for every $n \geq N$.

2.b

weight: 0.3 **(In class.)** Give an example in which Fatou's inequality is strict, even though every f_n has the same finite integral.

3. (Measures defined by non-negative densities)

Let $(X, \mathcal{A}, \mu)$ be a measure space, and let $h \colon X \to [0, +\infty]$ be measurable. Do *not* assume that $h \in L^1(\mu)$. For $E \in \mathcal{A}$, define

$$\nu(E) \coloneqq \int_E h \, \mathrm{d}\mu.$$

3.a

weight: 0.4 Prove that ν is a measure on $(X, \mathcal{A})$. Indicate precisely which integral limit theorems you use.

3.b

weight: 0.4 **(In class.)** Prove that for every non-negative measurable φ,

$$\int_X \varphi \, \mathrm{d}\nu = \int_X \varphi h \, \mathrm{d}\mu.$$

First prove the identity for indicator functions, then for non-negative simple functions, and finally for arbitrary non-negative measurable functions.

3.c

weight: 0.2 Prove that ν is finite if and only if $h \in L^1(\mu)$. If μ is σ-finite, prove in addition that ν is σ-finite if and only if $h < +\infty$ almost everywhere with respect to μ.

4. (Capturing an integral on sets of finite measure)

Let $(X, \mathcal{A}, \mu)$ be a measure space, and let $f\colon X \to [0, +\infty]$ be measurable.

4.a

weight: 0.6 Assume that $\int_X f \, \mathrm{d}\mu < +\infty$.

Prove that there is an increasing sequence $(E_n)_{n\in\mathbb{N}}$ of measurable sets such that

$$\mu(E_n) < +\infty \qquad \text{for every } n,$$

and

$$\int_{E_n} f \, \mathrm{d}\mu \uparrow \int_X f \, \mathrm{d}\mu.$$

Thus a finite integral can be captured by an increasing sequence of sets of finite measure, even when the ambient measure space is not σ-finite.

> **Free hint.** Define
> $$E_n \coloneqq \{x \in X\colon f(x) \geq 1/n\}.$$
> Use Markov's inequality to prove that every E_n has finite measure, and determine the union of the sets E_n.

4.b

weight: 0.4 Prove that, independently of whether $\int_X f \, \mathrm{d}\mu$ is finite or infinite, if $(E_n)_{n\in\mathbb{N}}$ is a sequence of measurable sets such that

$$E_n \uparrow X,$$

then

$$\int_{E_n} f \, \mathrm{d}\mu \uparrow \int_X f \, \mathrm{d}\mu.$$

Then show that if $(X, \mathcal{A}, \mu)$ is σ-finite, there exist measurable sets $(E_n)_{n\in\mathbb{N}}$ such that

$$E_n \uparrow X,$$

$$\mu(E_n) < +\infty \qquad \text{for every } n,$$

and

$$\int_{E_n} f \, \mathrm{d}\mu \uparrow \int_X f \, \mathrm{d}\mu.$$

In particular, the last limit is $+\infty$ when $\int_X f \, \mathrm{d}\mu = +\infty$.